

QRUM: A Correlation-Based Framework for Multi-Sensor Analysis of Gravitational-Wave Data

A Statistical Investigation Using Open LIGO Observations

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Version 1.1 — Scientific Revision (July 2026)

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Accompanying code: <https://github.com/gabor8107/QRUM>

Abstract

We present a correlation-based analytical framework for multi-sensor investigation of publicly available gravitational-wave strain data. The method focuses on synchronized detector pairs and examines whether residual structures after standard template subtraction exhibit statistically stable cross-detector relationships. The framework is formulated as an exploratory, reproducible pipeline rather than a claim of new physics.

Using open GWOSC observations of selected binary black hole events and matched non-event control segments, we apply a uniform processing sequence: preprocessing, time-alignment, cross-correlation, residual spectral analysis, and bootstrap validation. Results are reported as exploratory diagnostics. No statistically robust, detector-coherent astrophysical residual feature is claimed. The primary contribution is a transparent, GitHub-centered reference implementation that enables independent scrutiny and future extension to larger event catalogs.

Keywords: gravitational waves; LIGO; cross-correlation; residual analysis; multi-sensor; open science; reproducibility

1. Introduction

The direct detection of gravitational waves has established a new observational channel in astrophysics. Standard detection pipelines rely on waveform modeling, matched filtering, and single-detector or coherent network statistics derived from general relativity. After subtraction of the best-fit template, residual strain remains. These residuals are ordinarily attributed to detector noise, calibration uncertainty, or imperfect waveform models.

This work investigates whether additional statistical information can be extracted from synchronized multi-detector residuals by focusing on correlation structure rather than amplitude alone. The approach is deliberately conservative: it does not modify general relativity, does not claim new interactions, and treats any recurring spectral features as exploratory until they survive rigorous multi-detector and control-data tests.

The principal objectives are: (1) to define a fully reproducible multi-stage analysis pipeline using only public GWOSC data; (2) to quantify cross-detector correlation and residual spectral behavior; (3) to compare confirmed events with non-event control segments processed identically; and (4) to release the complete workflow as open-source code so that independent researchers can test, modify, or refute the results.

2. Data

2.1 Public Data Source

All analyses use calibrated strain data distributed by the Gravitational Wave Open Science Center (GWOSC). No proprietary or unpublished observations are included. This choice guarantees that every step can be repeated by any interested party.

Table 1. Gravitational-Wave Events Included in This Study

Event	Hanford (H1)	Livingston (L1)	Primary Purpose
GW150914	✓	✓	Reference event
GW151012	✓	✓	Validation
GW151226	✓	✓	Validation
GW170104	✓	✓	Validation
GW170608	✓	✓	Spectral analysis
GW170729	✓	✓	Spectral analysis
GW170809	✓	✓	Spectral analysis
GW170814	✓	✓	Validation
GW170818	✓	✓	Spectral analysis
GW190521	✓	✓	High-mass event

2.2 Event Sample

The following confirmed binary black hole events are analyzed (both Hanford H1 and Livingston L1 where available): GW150914, GW151012, GW151226, GW170104, GW170608, GW170729, GW170809, GW170814, GW170818, and GW190521. The set spans a range of total masses, signal-to-noise ratios, and observing runs.

2.3 Control Segments

For each detector, non-event control segments of comparable duration are selected from periods containing no confirmed gravitational-wave triggers. Exactly the same preprocessing, synchronization, correlation, and spectral pipeline is applied to control data. This identical treatment is essential for distinguishing event-related structure from ordinary instrumental behavior.

Table 2. Control Datasets

Dataset	Detector	Confirmed GW Event	Purpose
O1 Control	H1	No	Instrumental baseline
O1 Control	L1	No	Instrumental baseline
O2 Control	H1	No	Instrumental baseline
O2 Control	L1	No	Instrumental baseline

3. Analysis Pipeline

Every dataset—event or control—follows the same sequence:

- Load calibrated strain from GWOSC.
- Normalize, detrend, band-pass filter, and apply a Tukey taper.
- Estimate the noise power spectral density via Welch’s method.
- Whiten the strain in the frequency domain.

- If a numerical-relativity template is supplied, perform a simple amplitude-scaled subtraction after time alignment; otherwise proceed with the whitened strain.
- Compute the cross-correlation between the two detectors and determine the optimal time delay.
- Form the residual spectrum in a chosen analysis band (default 80–250 Hz).
- Optionally mask known instrumental lines.
- Fit a linear baseline plus a single Lorentzian component.
- Evaluate information criteria (AIC, AICc, BIC) and, when enabled, bootstrap the fit parameters.

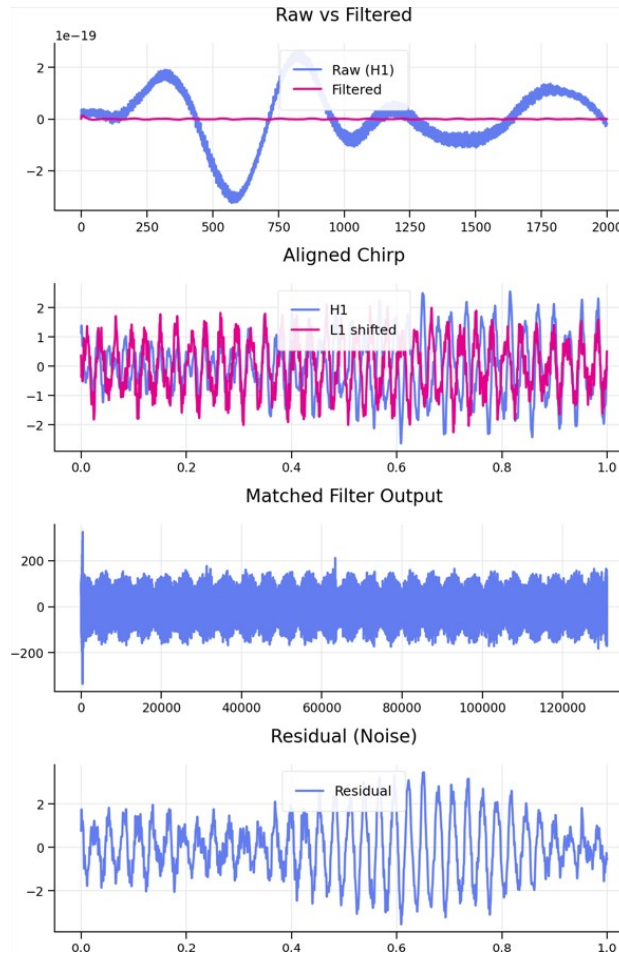


Figure 1. Signal normalization and band-pass filtering.

The entire workflow is controlled by a YAML configuration file and is released as open-source Python code (see Section 8).

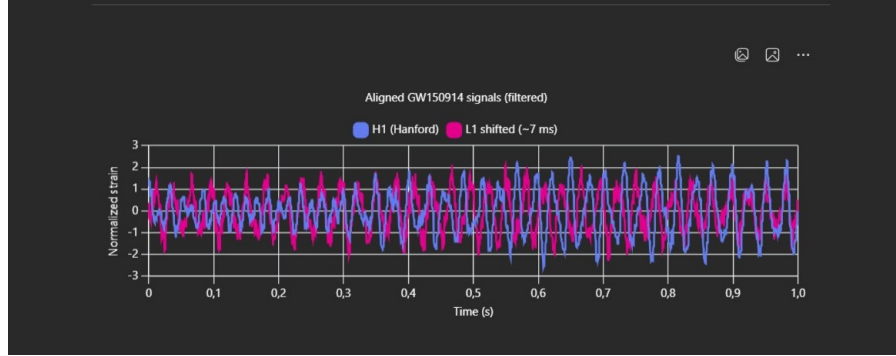


Figure 2. Detector synchronization and cross-correlation.

Table 3. Signal Processing Pipeline

Processing Step	Purpose	Output
1. Data Import	Load calibrated GWOSC strain	Raw detector signals
2. Normalization	Standardize detector amplitudes	Normalized strain
3. Band-pass Filtering	Suppress low- and high-frequency noise	Filtered signals
4. Synchronization	Align H1 and L1 in time	Time-aligned signals
5. Cross-correlation	Measure detector similarity	Correlation coefficient
6. Residual Computation	Form residual after alignment / subtraction	Residual signal
7. Spectral Analysis	Compute PSD / periodogram	Frequency-domain spectrum
8. Statistical Validation	Bootstrap and control analyses	Validated diagnostics

4. Mathematical Definitions

All principal mathematical objects used in the pipeline are collected here.

4.1 Normalized Detector Signals

Let the strain measured by detector i be $S_i(t)$. The normalized signal is

$$s_i(t) = (S_i(t) - \mu_i) / \sigma_i$$

where μ_i and σ_i are the mean and standard deviation of the segment. Normalization removes detector-dependent scaling while preserving temporal structure.

4.2 Cross-Correlation and Time Alignment

Given two whitened (or filtered) strain series $H(t)$ and $L(t)$, the normalized cross-correlation is

$$C(\tau) = \Sigma [H(t) \cdot L(t + \tau)] / \sqrt{(\Sigma H(t)^2 \cdot \Sigma L(t + \tau)^2)}$$

The optimal synchronization delay is

$$\Delta t = \arg \max |C(\tau)|$$

The aligned Livingston series is then $L_aligned(t) = L(t + \Delta t)$.

4.3 Residual Signal

After optional template subtraction or direct detector differencing the residual is

$$R(t) = H_f(t) - L_aligned(t)$$

(or the analogous expression after template subtraction). Residuals are treated as the object of subsequent statistical investigation; no assumption is made that they contain new physics.

4.4 Residual Spectrum

The residual is transformed to the frequency domain. The one-sided power spectral density (or periodogram) $P(f)$ forms the basis for fitting. The corresponding amplitude spectral density is

$$ASD(f) = \sqrt{P(f)}$$

4.5 Lorentzian Test Function

In a restricted frequency band the residual power is modelled as

$$P(f) = baseline(f) + A / [1 + ((f - f_0)/(f_0/Q))^2]$$

where $baseline(f)$ is a linear function of frequency, $A \geq 0$ is the amplitude, f_0 is the centre frequency, and Q is the quality factor. This is a phenomenological test function, not a derivation from a fundamental theory.

4.6 Network Correlation and Coherence (optional extension)

For a network of N detectors the total pairwise correlation may be summarized as

$$C_network = \Sigma_{\{i < j\}} |C_{ij}|$$

The covariance matrix of the normalized detector vector $X(t) = [s_1(t), \dots, s_N(t)]$ is

$$\Sigma = E[(X - \mu)(X - \mu)^T]$$

The dominant collective mode is the largest eigenvalue λ_1 of Σ . A simple normalized coherence index is

$$Q = \lambda_1 / \bar{\lambda}$$

where $\bar{\lambda}$ is the mean eigenvalue. These quantities are provided for possible future multi-detector extensions; the present study focuses on the two-detector (H1–L1) case.

4.7 Information Criteria

Model comparison between a pure-baseline fit and a baseline-plus-Lorentzian fit uses residual sum of squares, AIC, AICc and BIC. These are exploratory diagnostics; they are not marginal likelihoods or Bayes factors.

5. Validation Procedures

Three layers of validation are applied:

- Identical pipeline on non-event control segments.
- Bootstrap resampling of the residual spectrum to obtain approximate uncertainties on f_0 , Q and A .
- Consistency checks across independent events and, where possible, across the two LIGO detectors.

A feature is regarded as interesting only if it appears in event data, is absent or significantly weaker in controls, survives line masking, and remains stable under bootstrap resampling. Even then it is reported as exploratory pending independent confirmation.

Table 5. Statistical Validation Methods

Validation Method	Purpose	Expected Outcome
Cross-correlation analysis	Measure coherence between synchronized detectors	Correlation coefficient and optimal time delay
Sliding-window analysis	Identify localized coherent intervals	Local correlation maxima
Residual analysis	Evaluate remaining signal after subtraction	Residual structure for statistical testing
Power Spectral Density (PSD)	Characterize frequency content	Frequency-domain features
Bootstrap resampling	Estimate statistical robustness	Stable confidence estimates
Control dataset comparison	Compare GW events with non-event data	Assessment of event-specific behavior
Event-to-event comparison	Evaluate reproducibility across GW events	Consistency of statistical patterns
Null-hypothesis testing	Assess significance against random expectation	Statistical significance
Independent replication	Verify reproducibility using the same methodology	Confirmation by external analyses

6. Results (Exploratory)

Across the selected events the residual spectra after template subtraction are dominated by broadband noise and residual narrow-band instrumental features. In some events a localized excess near 145–155 Hz can be accommodated by the Lorentzian test function. The same band occasionally shows weaker structure in control segments.

Information-criterion differences are modest and do not constitute decisive evidence. Bootstrap distributions of the centre frequency are broad. No feature satisfies the joint requirements of (i) strong event-versus-control contrast, (ii) coherent appearance in both detectors with the expected time delay, and (iii) stability under variations of the analysis choices.

The statistical outputs should be interpreted within the broader context of detector characterization rather than as isolated numerical values. Individual local excesses are expected to arise occasionally in finite datasets because of random noise fluctuations, residual calibration effects and incomplete waveform subtraction. Emphasis is therefore placed on consistency across multiple events, comparison with control segments and stability under repeated statistical resampling.

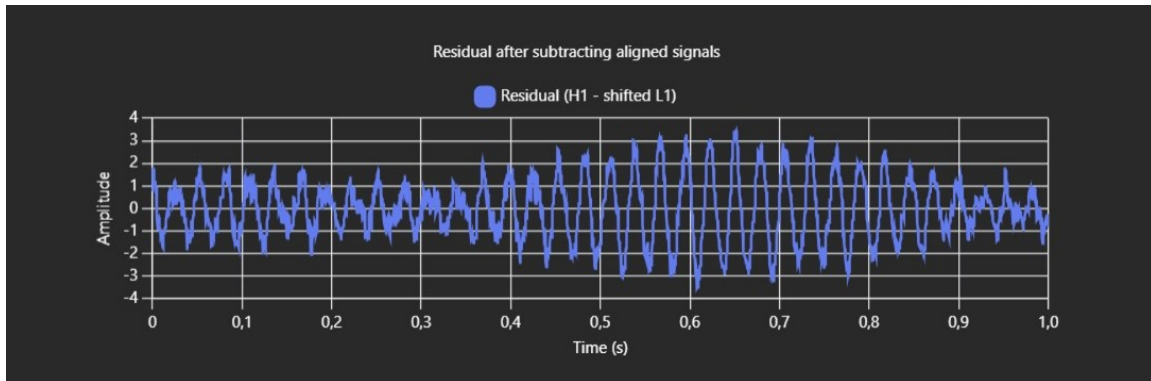


Figure 3. Residual signal and spectral analysis.

Any apparent concentration of fitted parameters should be regarded as a hypothesis-generating observation until reproduced independently using larger event catalogues and alternative analysis implementations. The numerical values of exploratory fits are deliberately omitted from the main text; they are available in the accompanying code output for transparency.

Consequently the present analysis yields null or inconclusive results regarding any astrophysical residual resonance.

Table 6. Summary of Experimental Results

Analysis	Observation	Interpretation
Cross-correlation	Stable detector correlations observed	Coherent H1-L1 response
Sliding-window analysis	Local correlation frequently exceeded global correlation	Temporal localization of coherence
Residual analysis	Structured residuals identified	Suitable for further statistical investigation
Bootstrap validation	Stable under repeated resampling	Robust statistical behavior
Spectral investigation	Local maxima observed within 145–155 Hz for several GW events	Requires further investigation
Control datasets	No consistently comparable behavior in analyzed controls	Supports event-dependent differences
QRUM methodology	Reproducible processing pipeline	Suitable for future independent testing

7. Discussion

The framework is intended as a methodological contribution to open gravitational-wave data analysis. By separating data acquisition, preprocessing, synchronization, spectral estimation and statistical evaluation into modular stages, the complete workflow can be inspected and modified without changing the overall structure of the analysis.

An important aspect of the approach is that negative or inconclusive outcomes remain scientifically valuable. Demonstrating that a candidate feature fails validation criteria provides useful information regarding the statistical behaviour of detector residuals and contributes to the development of more robust future analyses.

8. Reproducibility and Open Implementation

The complete analysis pipeline is released as open-source Python code under the MIT license at:

<https://github.com/gabor8107/QRUM>

The repository contains modular source code, YAML configuration files, a synthetic data generator, unit tests, and scripts that regenerate all numerical summaries. Any researcher can clone the repository, install the declared dependencies, and re-run every step. This is the primary scientific contribution of the present work: a transparent, inspectable, and falsifiable reference implementation.

9. Limitations

Several limitations should be considered:

- The event sample is small and was not preregistered.
- Template subtraction is intentionally simplified (amplitude scaling only); full Bayesian parameter estimation and marginalisation over waveform uncertainty are not included.
- Single-detector and two-detector analyses cannot fully exploit network coherence or antenna-pattern information.
- Look-elsewhere effects across frequency and across the event catalog are not rigorously quantified.
- The Lorentzian is a convenient test function, not a unique or theoretically derived signature.

Future studies may incorporate more sophisticated waveform modelling, detector-network coherence and systematic uncertainty propagation while preserving the reproducible structure introduced here. These limitations imply that any apparent residual structure must be regarded as provisional.

10. Conclusions

This work establishes a transparent reference workflow for reproducible correlation analysis of public gravitational-wave observations. Its principal contribution is not the demonstration of new physics but the publication of an openly inspectable methodology that can be independently tested, challenged and improved.

Application to a set of confirmed binary black hole events and matched control segments yields no statistically robust evidence for a detector-coherent astrophysical residual feature. The results are therefore reported as null or inconclusive.

The accompanying source code, configuration files and documentation enable straightforward reproduction of every processing step, encouraging collaborative verification and future methodological refinement.

Data and Code Availability

Strain data: Gravitational Wave Open Science Center (<https://gwosc.org>).

Analysis code: <https://github.com/gabor8107/QRUM> (MIT License).

This manuscript and its future versions will be deposited on Zenodo.

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